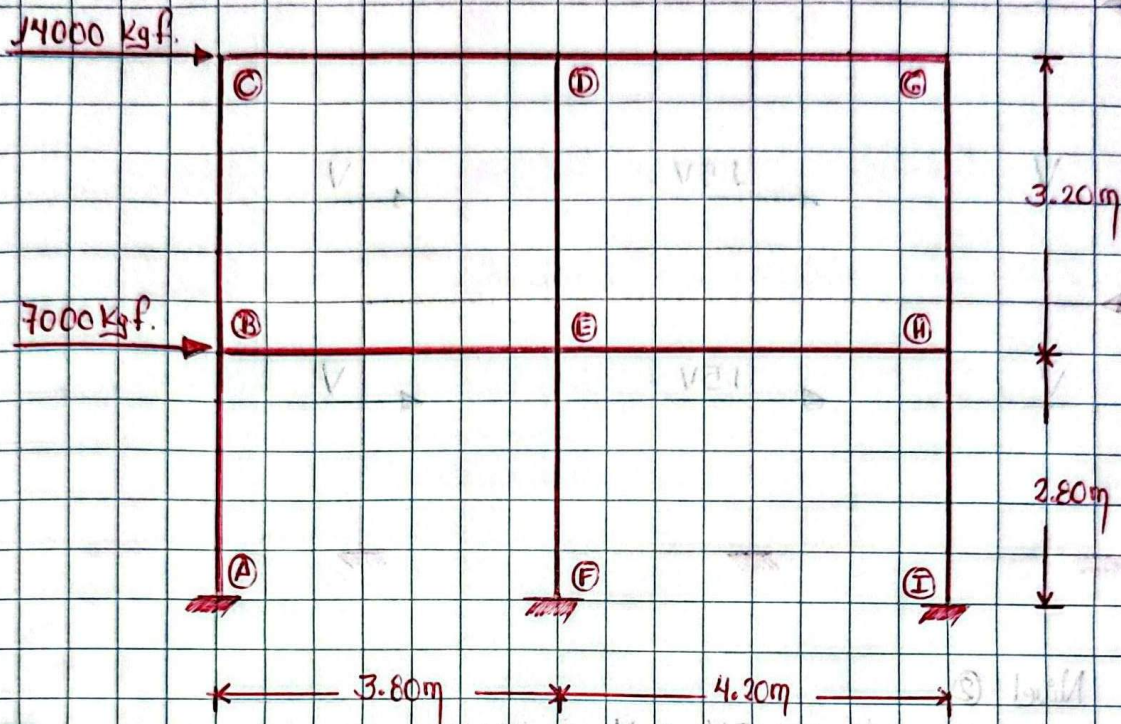
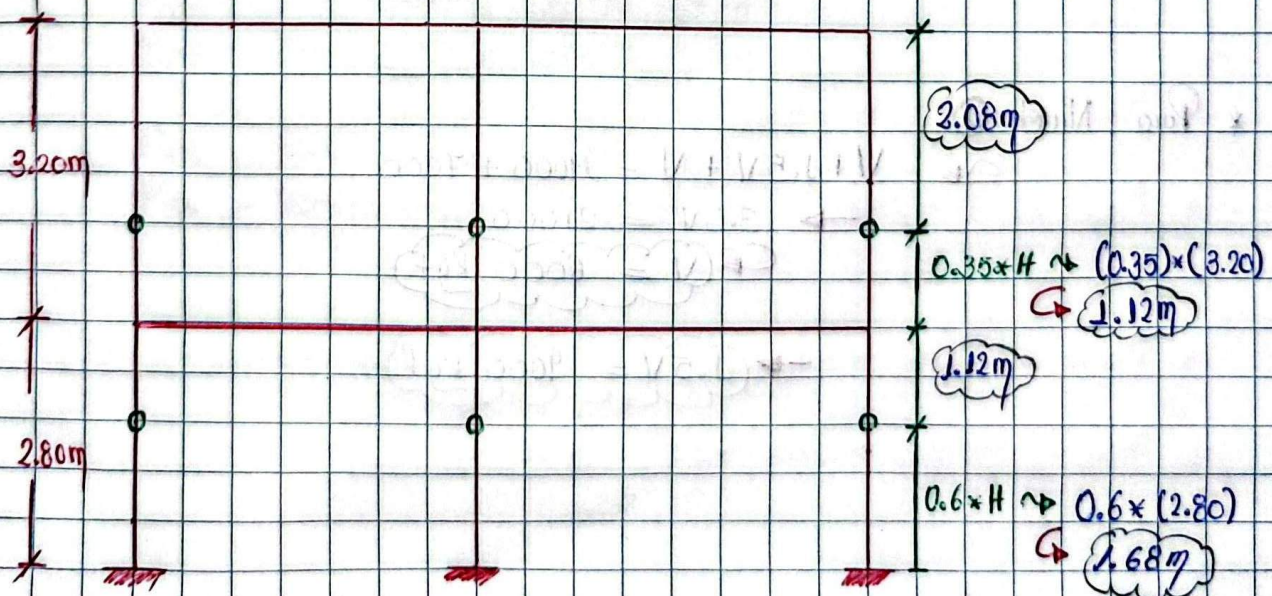


Ejercicio: Resolver el Portico usando el Método del Portal. -2

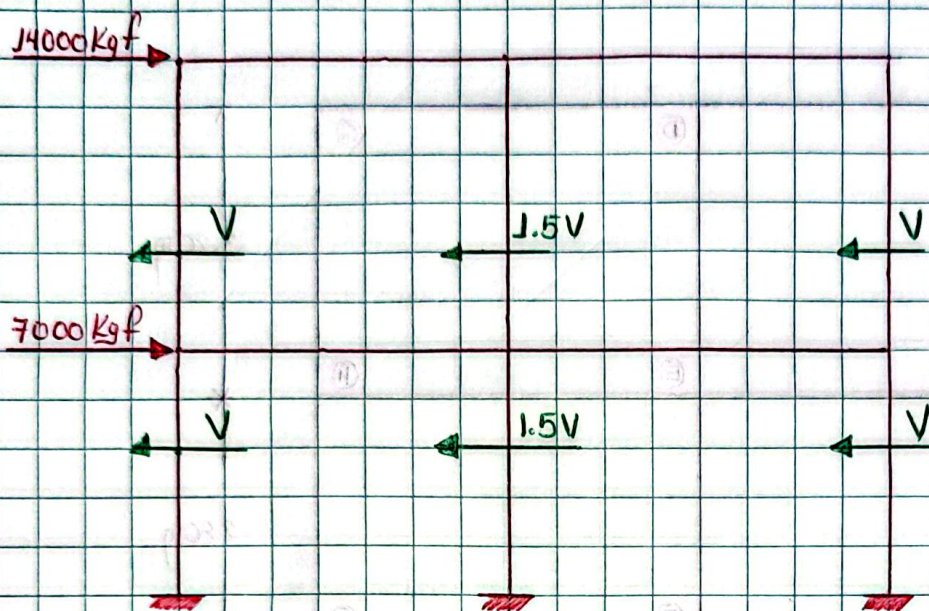


Solución:

1º Determinamos el punto de inflexión de las columnas.



2º Determinar las cortantes para cada columna.



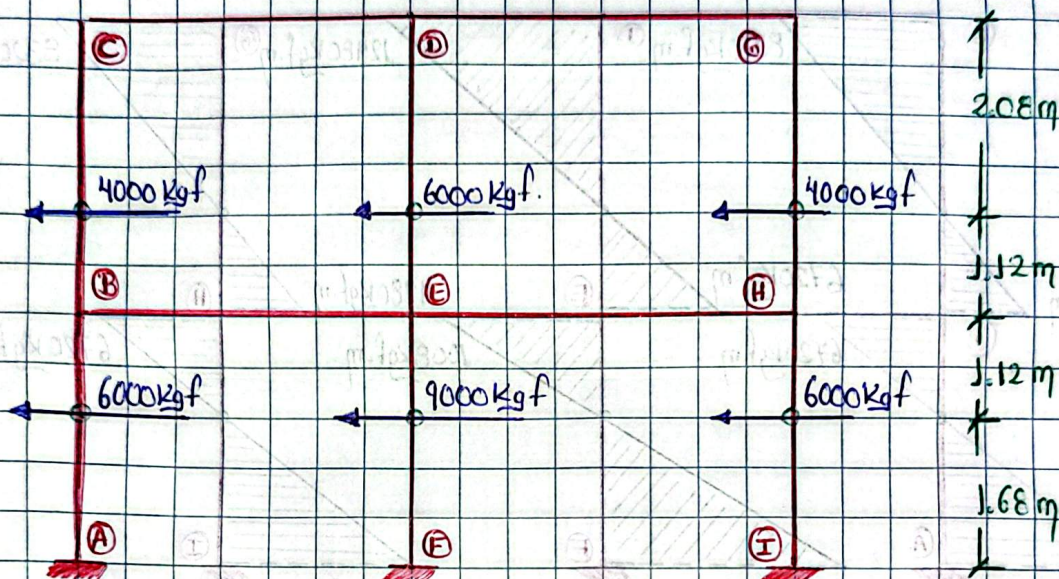
* Para Nivel ②

$$\begin{aligned} \rightarrow V + 1.5V + V &= 14000 \\ \rightarrow 3.5V &= 14000 \\ \rightarrow V &= 4000 \text{ Kg}f \\ \rightarrow 1.5V &= 6000 \text{ Kg}f \end{aligned}$$

* Para Nivel ①

$$\begin{aligned} \rightarrow V + 1.5V + V &= 14000 + 7000 \\ \rightarrow 3.5V &= 21000 \\ \rightarrow V &= 6000 \text{ Kg}f \\ \rightarrow 1.5V &= 9000 \text{ Kg}f \end{aligned}$$

3^o Determinar los momentos en el Pie y Cabeza de las columnas...

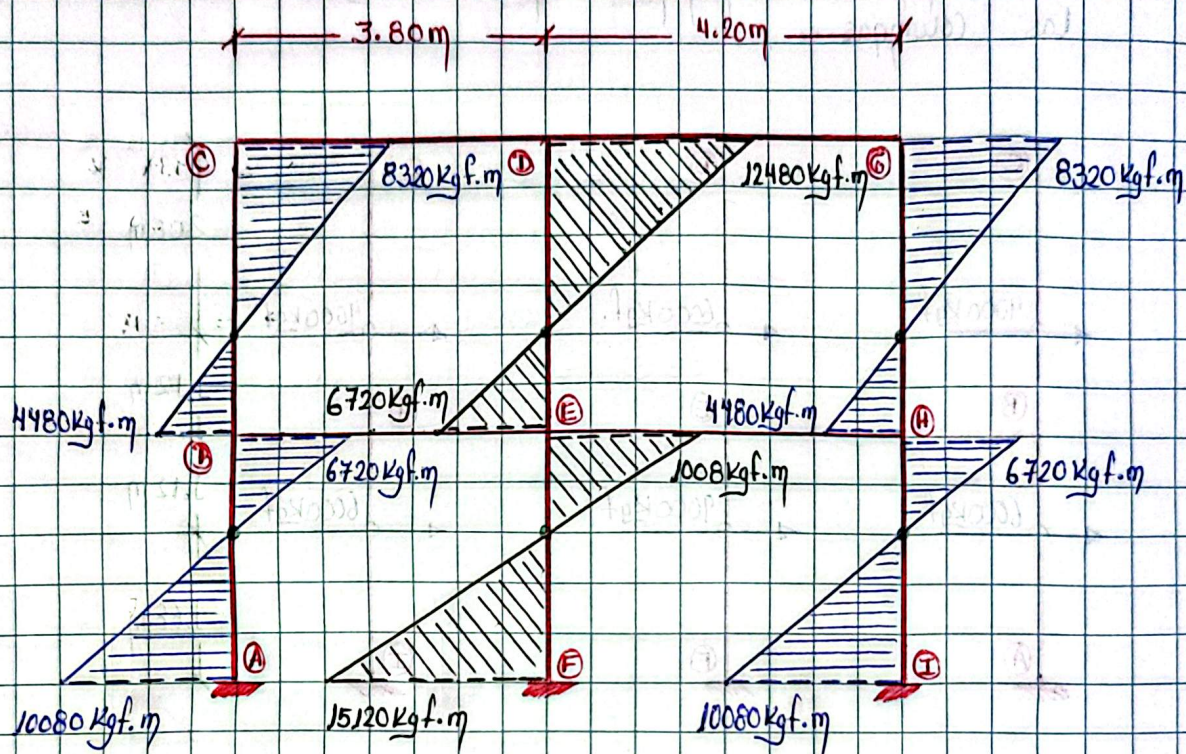


$$\begin{aligned} \curvearrowright M_{AB} &= (6000)(1.68) = 10080 \text{ kgf}\cdot\text{m} = M_{IH} \\ M_{BA} &= (6000)(1.12) = 6720 \text{ kgf}\cdot\text{m} = M_{HI} \end{aligned}$$

$$\begin{aligned} M_{BC} &= (4000)(1.12) = 4480 \text{ kgf}\cdot\text{m} = M_{HG} \\ M_{CB} &= (4000)(2.08) = 8320 \text{ kgf}\cdot\text{m} = M_{GH} \end{aligned}$$

$$\begin{aligned} M_{FE} &= (9000)(1.68) = 15120 \text{ kgf}\cdot\text{m} \\ M_{EF} &= (9000)(1.12) = 10080 \text{ kgf}\cdot\text{m} \end{aligned}$$

$$\begin{aligned} M_{ED} &= (6000)(1.12) = 6720 \text{ kgf}\cdot\text{m} \\ M_{DE} &= (6000)(2.08) = 12480 \text{ kgf}\cdot\text{m} \end{aligned}$$



4º Determinamos los momentos en las vigas.

$$* M_{BE} = -(4480 - 6720) = 2240 \text{ kgf.m}$$

$$* M_{EB} = (6720 - 10080) \times \left(\frac{\frac{1}{3.8}}{\frac{1}{3.8} + \frac{1}{4.2}} \right) = -1764 \text{ kgf.m}$$

$$* M_{CD} = 8320 \text{ kgf.m}$$

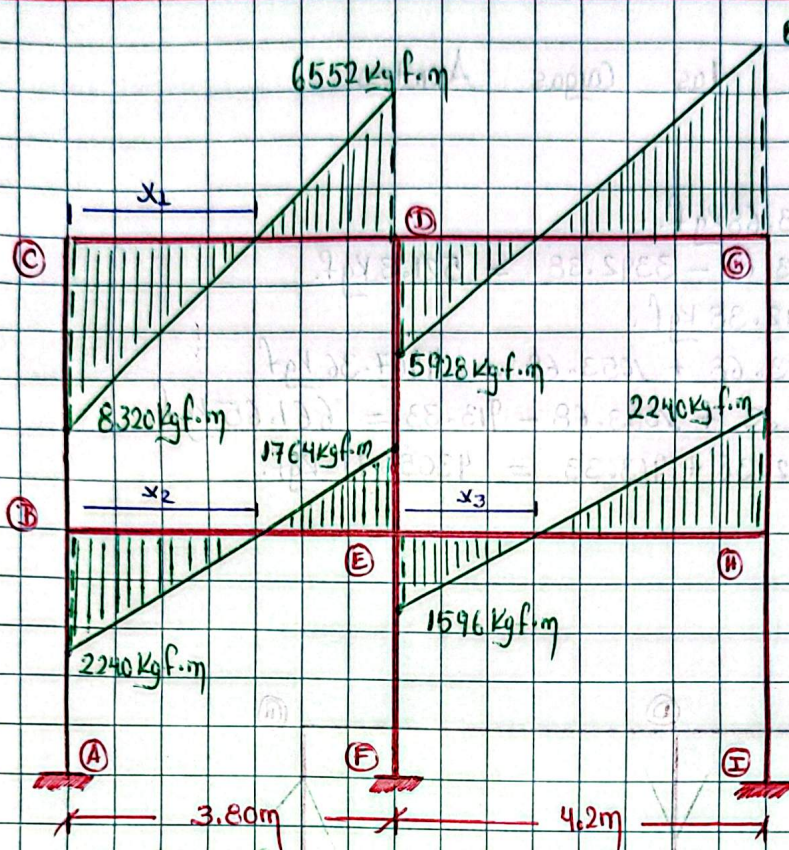
$$* M_{DC} = (0 - 12480) \times \left(\frac{\frac{1}{3.8}}{\frac{1}{3.8} + \frac{1}{4.2}} \right) = -6552 \text{ kgf.m}$$

$$* M_{EH} = (6720 - 10080) \times \left(\frac{\frac{1}{4.2}}{\frac{1}{3.8} + \frac{1}{4.2}} \right) = -1596 \text{ kgf.m}$$

$$* M_{HE} = -(4480 - 6720) = 2240 \text{ kgf.m}$$

$$* M_{DG} = (0 - 12480) \times \left(\frac{\frac{1}{4.2}}{\frac{1}{3.8} + \frac{1}{4.2}} \right) = -5928 \text{ kgf.m}$$

$$* M_{GD} = 8320 \text{ kgf.m}$$



$$\rightarrow x_1 = \frac{8320(3.8)}{8320 + 6552} = 2.13m$$

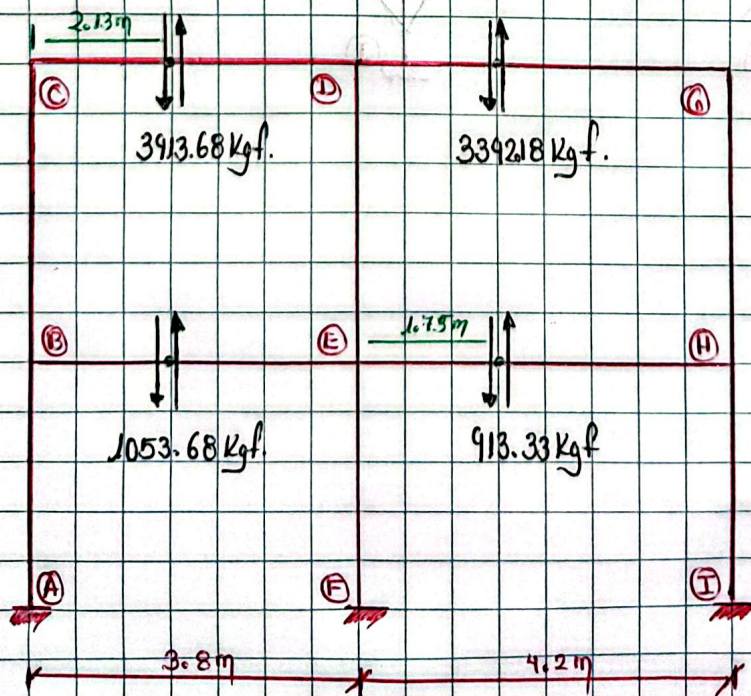
$$\rightarrow x_2 = 2.13$$

$$\rightarrow x_3 = 1.75$$

5° Determinamos las cortantes en las vigas.

$$* V_{BE} = \frac{2240 + 1764}{3.80} = 1053.68 \text{ kgf.}$$

$$* V_{CD} = \frac{8320 + 6552}{3.80} = 3913.68 \text{ kgf.}$$



$$* V_{EH} = \frac{1596 + 2240}{4.20} = 913.33 \text{ kgf.}$$

$$* V_{DG} = \frac{5928 + 8320}{4.20} = 3392.38 \text{ kgf.}$$

6º Determinamos las Cargas Axiales

- * $A_{CB} = 3913.68 \text{ Kg}.$
- * $A_{DE} = 3913.68 - 3392.38 = 521.3 \text{ Kg}.$
- * $A_{GH} = 3392.38 \text{ Kg}.$
- * $A_{BA} = 3913.68 + 1053.68 = 4967.36 \text{ Kg}.$
- * $A_{EF} = 521.3 + 1053.68 - 913.33 = 661.65 \text{ Kg}.$
- * $A_{HI} = 3392.38 + 913.33 = 4305.71 \text{ Kg}.$

